Singleron



01 Linux introduction

02 Linux system configuration

03 Linux common commands

04 Conda introduction & miniconda installation

05 Basic use of Conda

06 Appendix

0:36 / 16:38

What is Linux?

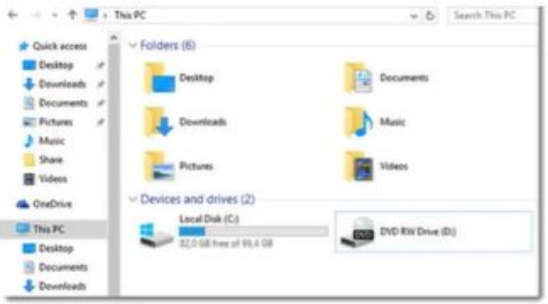
- Linux is a free and open source UNIX-like operating system whose kernel was first released by Linus Torvalds in 1991.
- As long as anyone follows the GNU General Public License (GPL), all of the underlying source code of Linux is freely available to any person or institution, as well as freely modified and redistributed.
- Support for multiple platforms.
- Supports multiple users and multiple tasks

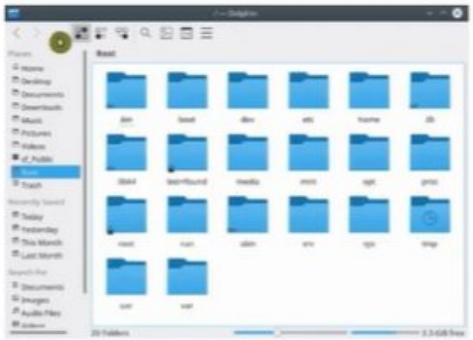
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- Support for multiple platforms.
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- **Differences between Linux and Windows**

Linux: root directory, character string
Windows: disk C, disk D, Pictures etc.





- **Windows**



(Mouse)
Single/double/right click

drag and drop

Enter

- **Linux**



Source

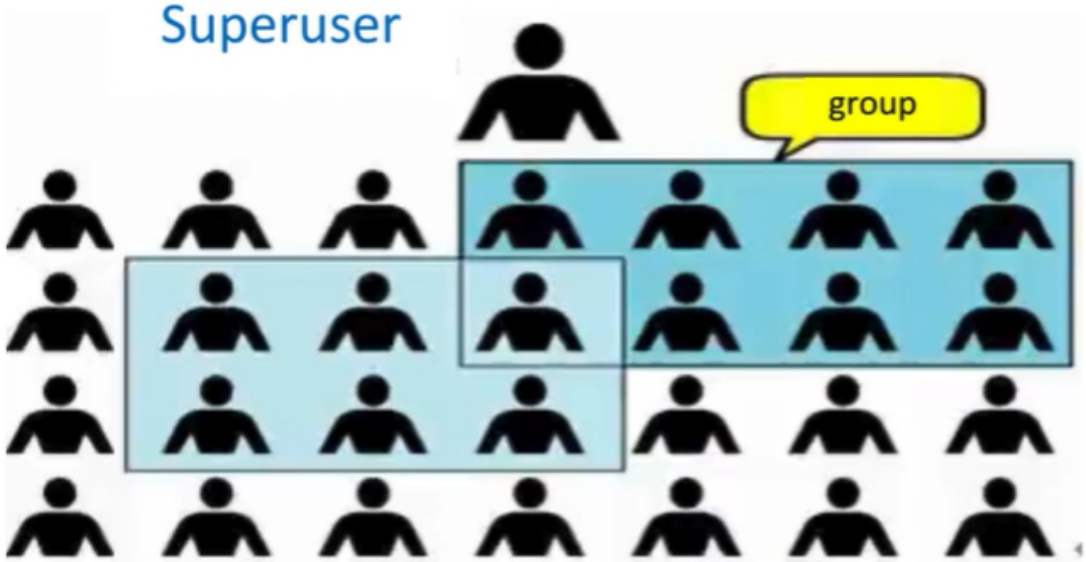
Help

Yes

cd/mv

Linux-users/groups

- **Users and group**



- Each process on the system is run by a particular user
- Each file is owned by a particular user
- Access to files and directories is limited by the user

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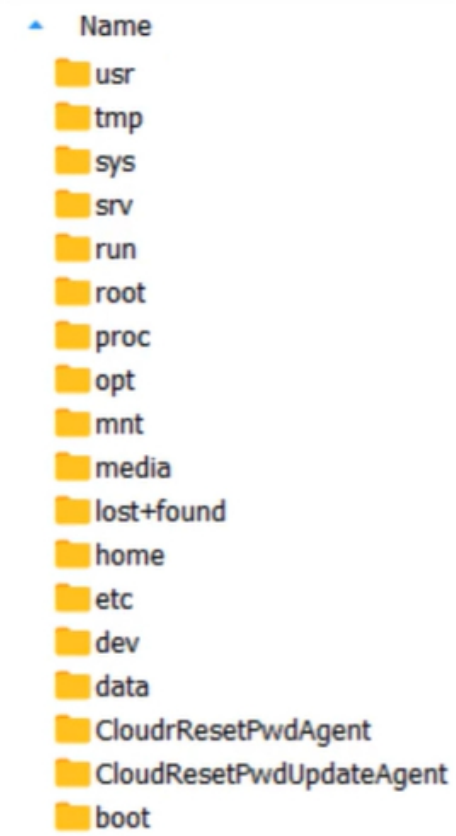
02 Linux System Configuration

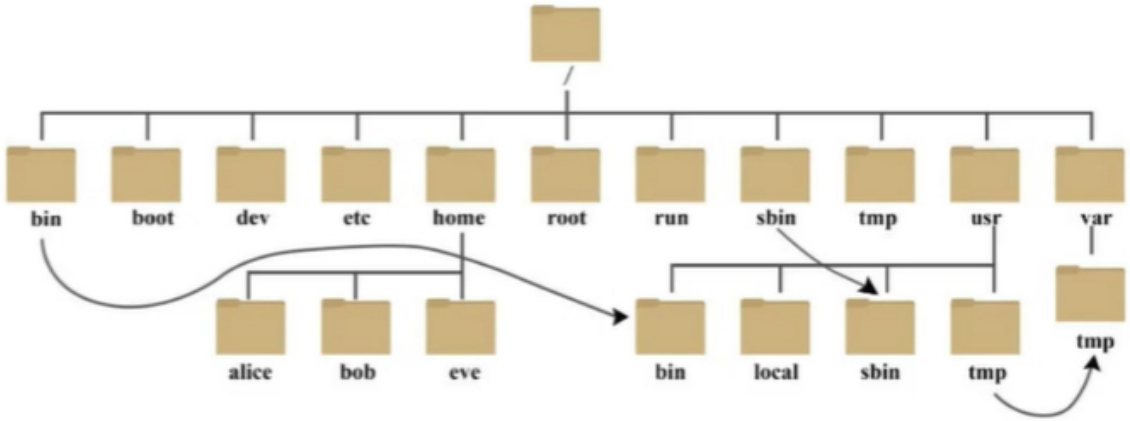


Linux system directory structure

- **Everything in Linux is a file**

/






```
root 15:36:46 /
$ ls
bin                       data  lib         mnt   run   tmp
boot                      dev   lib64       opt   sbin  usr
CloudResetPwdUpdateAgent  etc   lost+found  proc  srv   var
CloudrResetPwdAgent       home  media       root  sys
```

Absolute and relative paths

- Absolute path: The root directory (/), the name of the file or directory to start with, and the name of each directory you must pass through to find the object (directory or file). It is a complete guidepost to the location of the file, so you can use the absolute path to find the required file in any case.

```
Windrunner 21:21:51 ~/singleron/software
$ pwd
/mnt/sdd/singleron_lecturer/Windrunner/singleron/software
```

- Relative path: Not starting with a forward slash (/), starting from the current directory to the name of every directory you must pass through to find an object (directory or file).
  - '.' : indicates the current directory
  - '..' : indicates the upper-level directory
  - '~': indicates the home directory of the user.

```
Windrunner 21:24:46 /mnt/sdd/singleron_lecturer
$ cd ..
Windrunner 21:24:49 /mnt/sdd
$ cd ./singleron_lecturer/Windrunner/singleron/software
Windrunner 21:25:11 ~/singleron/software
```

Naming conventions for Linux files and directories

- All characters except the slash are valid. If special characters are included, enclose them in quotation marks.
- The file name or directory name contains a maximum of 256 characters
- Sensitive to upper- and lower-case letters.
  Singleron，singleron
- Avoid using plus or minus signs or “ . ”as the first character of a normal file name.
- The file type is obtained by the header information of the file, and the file type is not distinguished by the suffix. (Unlike Windows)

```
Windrunner 22:06:09 ~
$ ll
total 384
-rw-rw-r--   1 Windrunner Windrunner  68436 Apr  7 14:41 AA.png
drwxrwxr-x   3 Windrunner Windrunner   4096 Mar 18 09:55 accuracode_project
-rwxr--r--   2 Windrunner Windrunner    754 Mar 31 11:29 adduser_1.sh
lrwxrwxrwx   1 Windrunner Windrunner     10 Apr 24 16:39 adduser_2.sh -> adduser.sh
-rwxr--r--   2 Windrunner Windrunner    754 Mar 31 11:29 adduser.sh
drwxrwxr-x   2 Windrunner Windrunner   4096 Mar 31 12:59 ArchR
drwxrwxr-x   2 Windrunner Windrunner   4096 Apr 24 16:18 biosoftware
drwxrwxr-x   2 Windrunner Windrunner   4096 Apr 19 10:12 downstream_data_analysis
drwxrwxr-x   2 Windrunner Windrunner   4096 Apr  5 21:37 list_diffgene
drwxrwxr-x  16 Windrunner Windrunner   4096 Mar 22 22:32 miniconda3
-rwxrwxr-x   1 Windrunner root       217660 Apr  1 20:52 nohup.out
drwxrwxr-x   3 Windrunner Windrunner   4096 Apr 24 10:37 practice
drwxrwxr-x   3 Windrunner Windrunner   4096 Mar 31 09:55 project
```

- Common file types:
  1. Regular file (-rwxrwxrwx)
  2. Directory (drwxrwxrwx)
  3. Symbolic link (lrwxrwxrwx)
  4. Block device file (brwxrwxrwx)
  5. Character device file (crwxrwxrwx)
  6. Pipe (prwxrwxrwx)
  7. Socket (srwxrwxrwx)

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03 Common Linux Commands



Common Linux Commands



Common Linux Commands

- File and directory management
- Creating, deleting, moving and copying files or directories
- File query
- View file contents
- Create links
- Task submission

Basic format of Linux commands

command + options + arguments → **ls –a /mnt/sdd/singleron_lecturer/Windrunner**

- A command usually consists of three parts:

  command name + command options + arguments (file name)
- There are Spaces between commands and arguments. Some commands must be followed by arguments, while others do not (with default arguments).

ls

- -a： Displays all files, including hidden files
- -l： Display Details
- -d： Viewing Directory Properties
- arguments： Refers to the object of the command.

```
Windrunner 14:57:54 ~
$ ls -l /mnt/sdd/singleron_lecturer/Windrunner
total 137076
-rw-rw-r--  1 Windrunner Windrunner    68436 Apr  7 14:41 AA.png
drwxrwxr-x  3 Windrunner Windrunner     4096 Mar 18 09:55 accuracode_project
lrwxrwxrwx  1 Windrunner Windrunner       10 Apr  7 15:01 adduser_2.sh -> adduser.sh
-rwxr--r--  1 Windrunner Windrunner      754 Mar 31 11:29 adduser.sh
drwxrwxr-x  2 Windrunner Windrunner     4096 Mar 31 12:59 ArchR
drwxrwxr-x  2 Windrunner Windrunner     4096 Apr  3 19:41 biosoftware
drwxrwxr-x  4 Windrunner Windrunner     4096 Apr  1 09:09 celescope_test_data
drwxrwxr-x  2 Windrunner Windrunner     4096 Apr  7 14:30 classnew
drwxrwxr-x  2 Windrunner Windrunner     4096 Apr 19 10:12 downstream_data_analysis
drwxrwxr-x  2 Windrunner Windrunner     4096 Apr  5 21:37 list_diffgene
drwxrwxr-x 16 Windrunner Windrunner     4096 Mar 22 22:32 miniconda3
-rw-rw-r--  1 Windrunner Windrunner    37510 Apr  7 15:05 newfile1.txt
-rwxrwxr-x  1 Windrunner root         217660 Apr  1 20:52 nohup.out
```

Lists directories and help commands

- **list directory contents: ls**

```
#Multiple column display
ls

#Detailed information is displayed in a single column
ls -l

#Show hidden files
ls -a
```

```
Windrunner 14:33:59 ~
$ ll
total 137076
-rw-rw-r--  1 Windrunner Windrunner    68436 Apr  7 14:41 AA.png
drwxrwxr-x  3 Windrunner Windrunner     4096 Mar 18 09:55 accuracode_project
lrwxrwxrwx  1 Windrunner Windrunner       10 Apr  7 15:01 adduser_2.sh -> adduser.sh
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-rw-rw-r--  1 Windrunner Windrunner    37510 Apr  7 15:05 newfile1.txt
-rwxrwxr-x  1 Windrunner root         217660 Apr  1 20:52 nohup.out
drwxrwxr-x  3 Windrunner Windrunner     4096 Apr  7 14:57 practice
drwxrwxr-x  3 Windrunner Windrunner     4096 Mar 31 09:55 project
```

- **Help Command Usage**

whatis: A brief description of a command.
which: Check the location of the executable file.
man: specifies the command parameters and how to use the command.
help: Get the help information of the shell's built-in commands.
Info: The function of the command is similar to that of the man command, and the information about the command is displayed.

```
(base) Windrunner 15:50:24 ~
$ which ls
alias ls='ls --color=auto'
        /usr/bin/ls
(base) Windrunner 15:50:30 ~
$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
```

5:59 / 16:38

- **O key exit, similar to vim:**

Change directory/copy/move or rename

Change working directory: cd

- cd is short for Change Directory, which is the command used to change the working directory.

```
#Switch to $home
cd ~

#Switch to the previous directory
cd -

#Switch to parent directory (upper level)
cd ..

#Switch to the current directory
cd .
```

Display the current directory command: pwd

- pwd is short for Print Working Directory, which is the command to display the directory you are currently in.

Copy: cp

```
#Copy the file to the directory
cp file folder

#Copy the file to the directory and rename it
Copy file folder/file2

#Copy directory
cp -r folder1 folder2

#Copy the soft connection, not the file
cp -d file1 file2
```

Move or rename: mv

```
##mv Move folder
#If folder2 already exists, move to folder2
#If folder2 does not exist, rename folder1 to folder2
mv folder1 folder2

##mv moving file
#If a folder already exists, you move it under a folder
#If a folder doesn't exist, rename file to folder
mv file folder
```

Creating a Directory

- create a directory : mkdir
  - -p : Create all directories recursively
  - -m : Manually sets the permission to create a directory

```
Windrunner 10:30:42 ~/classnew
$ ll
total 0
Windrunner 10:30:43 ~/classnew
$ mkdir class_new
Windrunner 10:31:12 ~/classnew
$ ls
class_new
```

```
(base) Windrunner 17:17:46 ~/singleron
$ mkdir -m 777 software
(base) Windrunner 17:18:01 ~/singleron
$ ll
total 8
drwxrwxr-x 3 Windrunner Windrunner 4096 Mar 30 16:56 project
drwxrwxrwx 2 Windrunner Windrunner 4096 Mar 30 17:18 software
```

```
Windrunner 16:55:51 ~/singleron
$ mkdir -p project/1/2/3/4
Windrunner 16:56:38 ~/singleron
$ ls
project
Windrunner 16:56:42 ~/singleron
$ tree
.
└── project
    └── 1
        └── 2
            └── 3
                └── 4

5 directories, 0 files
```

```
$ ls -l
total 4
drwxrwxr-x 3 Windrunner Windrunner 4096 Apr 24 10:37 project
```

View file contents

- **In Linux, you can use the following commands to view the contents of a file:**

- cat: Displays the contents of the file starting with the first line
- tac: Displays from the last line, you can see that tac is cat written backwards!
- nl: When displayed, print the line number!
- more: Displays the file content page by page
- less: Similar to more, but better; can turn the page forward!
- head: Just read the first few lines
- tail: Read only the last few lines of classes

Use man [command] to view the usage documents of each command. for example, man cat.

```
$ man cat
CAT(1)                                          User Commands
CAT(1)
NAME
       cat - concatenate files and print on the standard output
SYNOPSIS
       cat [OPTION]... [FILE]...
DESCRIPTION
       Concatenate FILE(s), or standard input, to standard output.
       -A, --show-all
              equivalent to -vET
       -b, --number-nonblank
              number nonempty output lines, overrides -n
       -e     equivalent to -vE
```

View file contents

- ## Viewing Commands: cat
  - The file contents are displayed from the first line.
  - Concatenate the file and print it to the standard output device.

```
-A, --show-all            equivalent to -vET
-b, --number-nonblank     number nonempty output lines, overrides -n
-e                        equivalent to -vE
-E, --show-ends           display $ at end of each line
-n, --number              number all output lines
-s, --squeeze-blank       suppress repeated empty output lines
-t                        equivalent to -vT
-T, --show-tabs           display TAB characters as ^I
```

```
#Display line number
cat -n file

#Merge files
cat file1 file2 file3 > file4

#Appends to the end of the file
cat file1 >> file2
```

```
Windrunner 14:03:15 ~/list_diffgene
$ ll
total 8
-rw-rw-r-- 1 Windrunner Windrunner 623 Mar 31 14:00 test_diffgenes1.xls
-rw-rw-r-- 1 Windrunner Windrunner 514 Mar 31 14:00 test_diffgenes2.xls
Windrunner 14:04:28 ~/list_diffgene
$ cat test_diffgenes1.xls test_diffgenes2.xls > test_diffgenes3.xls
Windrunner 14:05:05 ~/list_diffgene
$ ll
total 12
-rw-rw-r-- 1 Windrunner Windrunner  623 Mar 31 14:00 test_diffgenes1.xls
-rw-rw-r-- 1 Windrunner Windrunner  514 Mar 31 14:00 test_diffgenes2.xls
-rw-rw-r-- 1 Windrunner Windrunner 1137 Mar 31 14:05 test_diffgenes3.xls
```

7:21 / 16:38

View file contents

- **head**

-c <K>, print the first K bytes of each file
-n <K>, print the first K lines instead of the first 10
-q, never print headers giving file names
-v, always print headers giving file names

- **tail**

-c <K>,output the last K bytes
-n <K>, print the last K lines

```
Windrunner 15:04:04 ~
$ head -n 5 test_diffgenes.xls
gene_id p_val   avg_logFC       pct.1   pct.2   p_val_adj
FTH1    2.709528857114 47e-11   -0.592499992784135      1       1       5.76560645505388e-07
EIF1    1.57964538445031e-10    -0.527195749736074      0.989   0.993   3.36132741357181e-06
RPL10P9 2.85051172560295e-10    1.47314961280492        0.333   0.03    6.06560390091052e-06
B2M     6.10591489067906e-09    -0.344932349626 86      1       1       0.00012992776295876
```

Creating a File

- **Create file command：touch**

- 1. Used to update the time label of existing files to the current time of the system (default), their data will be preserved intact.
- 2. Used to create a new empty file.

```
-a                      change only the access time
-c, --no-create         do not create any files
-d, --date=STRING       parse STRING and use it instead of current time
-f                      (ignored)
-h, --no-dereference    affect each symbolic link instead of any referenced
                        file (useful only on systems that can change the
                        timestamps of a symlink)
-m                      change only the modification time
-r, --reference=FILE    use this file's times instead of current time
-t STAMP                use [[CC]YY]MMDDhhmm[.ss] instead of current time
```

```
(base) Windrunner 17:33:52 ~/singleron/test
$ touch test1.txt test2.txt
(base) Windrunner 17:33:58 ~/singleron/test
$ ll
total 0
-rw-rw-r-- 1 Windrunner Windrunner 0 Mar 30 17:33 test1.txt
-rw-rw-r-- 1 Windrunner Windrunner 0 Mar 30 17:33 test2.txt
(base) Windrunner 17:34:02 ~/singleron/test
$ touch -c test3.txt
(base) Windrunner 17:34:22 ~/singleron/test
$ ll
total 0
-rw-rw-r-- 1 Windrunner Windrunner 0 Mar 30 17:33 test1.txt
-rw-rw-r-- 1 Windrunner Windrunner 0 Mar 30 17:33 test2.txt
```

```
$ touch -t 201808080808 software_test1.txt
(base) Windrunner 17:39:49 ~/singleron/test
$ ll
total 0
-rw-rw-r-- 1 Windrunner Windrunner 0 Aug  8  2018 software_test1.txt
-rw-rw-r-- 1 Windrunner Windrunner 0 Mar 30 17:39 test1.txt
-rw-rw-r-- 1 Windrunner Windrunner 0 Mar 30 17:35 test3.txt
(base) Windrunner 17:39:50 ~/singleron/test
$ stat test1.txt
  File: test1.txt
  Size: 0               Blocks: 0          IO Block: 4096   regular empty file
Device: fd21h/64801d    Inode: 14164437    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1006/Windrunner)   Gid: ( 1006/Windrunner)
Access: 2022-03-30 17:39:34.173426345 +0800
Modify: 2022-03-30 17:39:34.173426345 +0800
Change: 2022-03-30 17:39:34.173426345 +0800
 Birth: -
```

Delete

- Remove a file or directory： rm

  - -f : force, ignore nonexistent files and arguments, never prompt;
  - -r： Recursive delete! Most commonly used in directory deletion! This is a very dangerous option!!
  - -i： prompt before every removal（Write .bashrc to prevent misoperation）
    alias rm='rm -i'
  - rmdir： Deleting an empty Directory

!!! There is no recycle bin in Linux !!!

Vim Text Editor

- The most common text editor on Linux and Unix systems
  - **vi/vim**
  - Has different modes

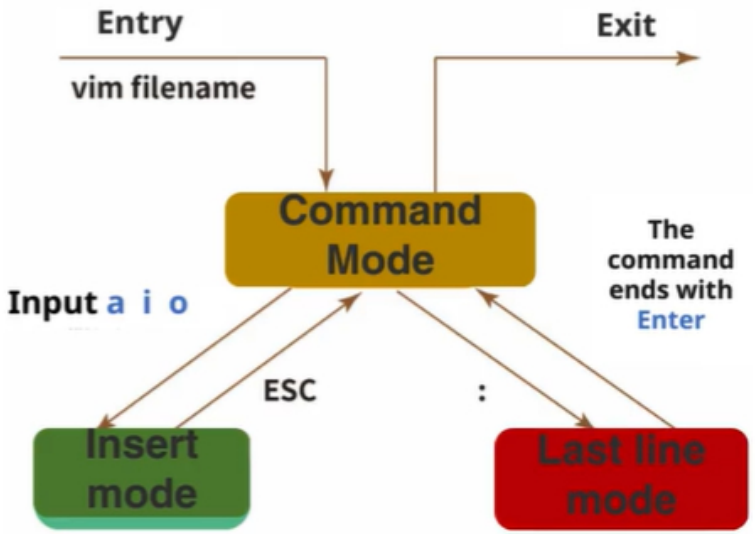
    1- Command Mode
- Move the cursor and use shortcut keys to copy, paste, or delete a file.

  2- Edit mode
- In command mode, enter a lowercase letter a or i to enter the editing mode. In this mode, you can edit the contents of the file. Esc to exit insert mode.

  3- Last line mode (Bottom line command mode)
- Enter the colon (:) in command mode to enter the last line mode. You can enter commands in the last line to search, replace, save, and exit files.

- x - to delete the unwanted character
- u - to undo the last the command and U to undo the whole line
- CTRL-R to redo
- A - to append text at the end
- :wq - to save and exit
- :q! - to trash all changes and exit
- dw - move the cursor to the beginning of the word to delete that word

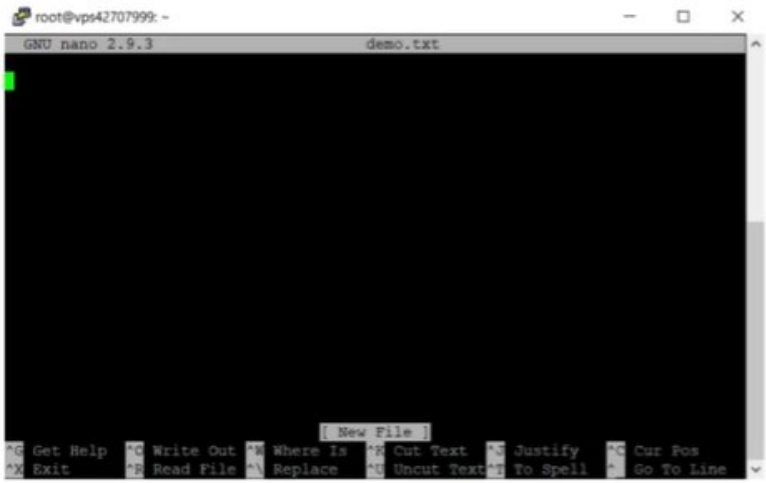
vi/vim working mode

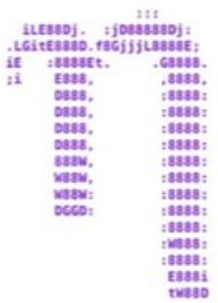


Cheat sheet: https://vim.rtorr.com/

Nano Text Editor

- **GNU nano** is a small and friendly text editor.
- Many features like an interactive search and replace, go to line and column number, auto-indentation, feature toggles, internationalization support, and filename tab completion.
- Unlike vi, nano is a modeless editor, which means that you can start typing and editing the text immediately after opening the file.





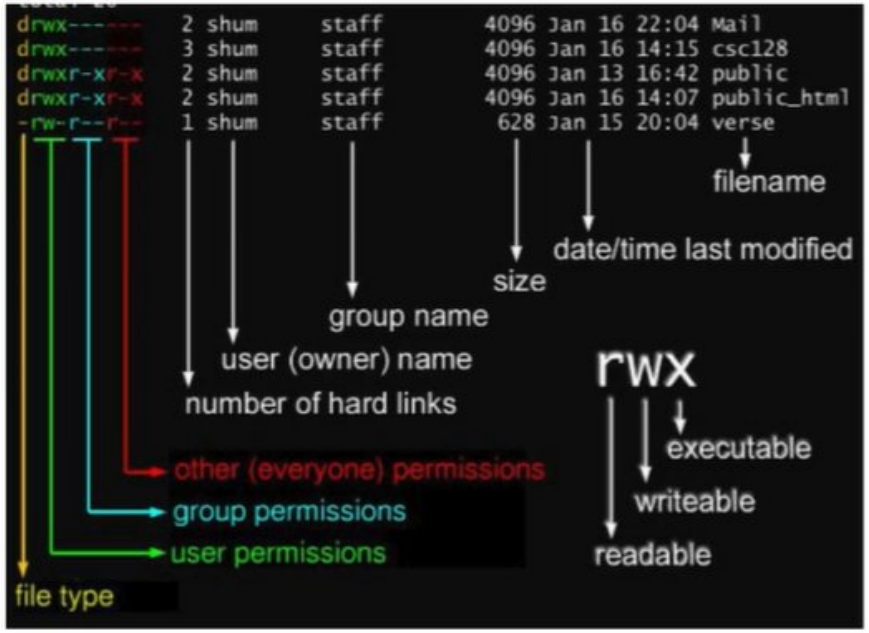
Command	Explanation
CTRL + A	Lets you jump to the beginning of the line.
CTRL + E	Lets you to jump to the end of the line.
CTRL + Y	Scrolls page down.
CTRL + V	Scrolls page up.
CTRL + G	A **Help** window will pop out and show you all the available commands.
CTRL + O	To save the file. Nano will ask you to edit or verify the desired file name.
CTRL + W	Search for a specified phrase in your text. Press **ALT + W** to search for the same phrase again.
CTRL + K	It cuts the entire selected line to the **cut buffer** (similar to clipboard).
CTRL + U	To paste the text from the cut buffer into the selected line.
CTRL + J	Justifies the current paragraph.
CTRL + C	Shows the current cursor position in the text (line/column/character).
CTRL + R	Opens a file and inserts it at the current cursor position.
CTRL + X	To exit Nano text editor. It prompts a save request if you made any changes to the file.

Cheat sheet: https://www.nano-editor.org/dist/latest/cheatsheet.html

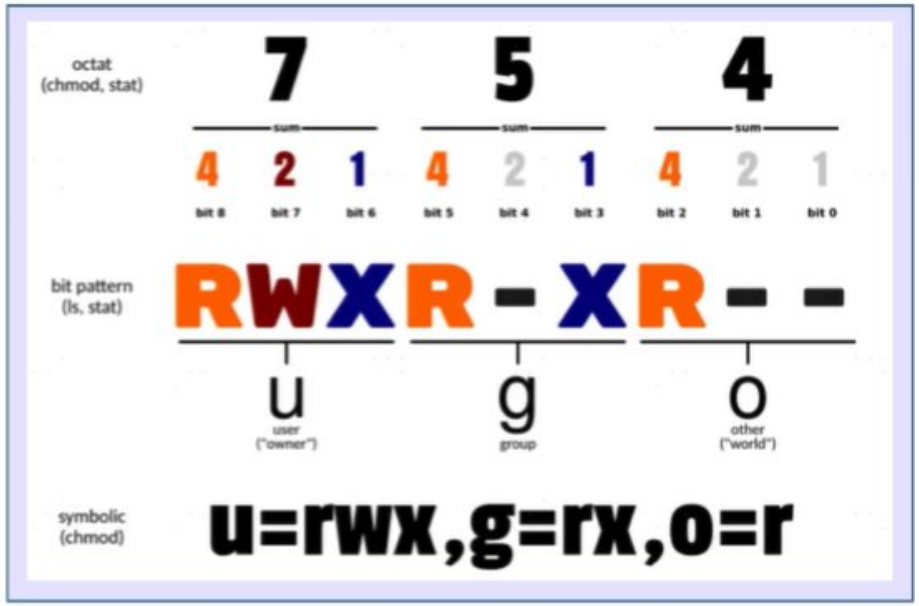
Permission Modification

File Directory Permission

- The Linux permission structure consists of three types of permissions:
  - the file owner, the group to which the owner belongs, and other group permissions.
- Each type has three permissions: read-only, writable, and execute.



- **owner (u)**
- **group (g)**
- **other (o)**
- **chown**
- **chgrp**
- **chmod**



- chmod modifies the permission:
  Prerequisite: Has the permission to operate on files (personal files)
  Grammar: chmod [{ugoa}{+-=}{rwx}] [file or directory]
  Usage: chmod [option] mode file/directory
  chmod 755 test.txt
  chmod –R 755 test

Within ec2

Package compress and uncompress

- **Package compression, decompression: tar**

- usage： tar [option] [file or directory name]

```
tar -cf archive.tar foo bar  # Create archive.tar from files foo and bar.
tar -tvf archive.tar         # List all files in archive.tar verbosely.
tar -xf archive.tar          # Extract all files from archive.tar.
```

```
Windrunner 22:10:08 ~/test
$ tar -cvf class.tar class/
class/
class/1/
class/1/2/
class/1/2/3/
class/1/2/3/4/
```

```
Windrunner 22:14:43 ~/test
$ tar -zcvf class.tar.gz class/
class/
class/1/
class/1/2/
class/1/2/3/
class/1/2/3/4/
Windrunner 22:14:59 ~/test
```

```
#Package files
tar -cf target.tar file1 file2 file3

#Extract tar to directory
tar -xf source.tar -C directory

#Package the file and gzip it
tar -czf target.tar.gz file1 file2 file3

#Decompress tar.gz
tar -xzf source.tar.gz

#Package the file and bzip2 compress it
tar -cjf target.tar.gz file1 file2 file3

#Decompress tar.gz2
tar -xjf source.tar.gz -C directory

#zip compression zip
zip -r compressed.zip directory

#Decompress the zip
unzip -r compressed.zip -d directory
```

Note: -f is immediately followed by the document name

Wildcard character

Wildcard character	meaning
*	Matches any string/text, including an empty string; * Represents any character (0 or more)
?	Matches any character (when not in parentheses)? Represents any 1 character
[abcd]	Matches any character in abcd
[a-z]	Indicates the range a to z. Indicates the meaning of the range. Matches any character in []
{..}	Represents the generated sequence. Delimited by commas and cannot contain Spaces
[!abcd]	Does not match any of the characters in the []

```
$ ll
total 8088
drwxr-xr-x. 2 wanghuimin ssh.randd     925 May 31 23:18 amulti
drwxr-x--x. 3 wanghuimin ssh.randd    1031 Aug 11 13:45 display
-rw-r--r--. 1 wanghuimin ssh.randd     274 May 31 13:42 readme.txt
drwxr-x--x. 2 wanghuimin ssh.randd     178 Jun  8 17:20 reports
-rw-r--r--. 1 wanghuimin ssh.randd 6787854 May 31 23:18 Rplots.pdf
-rwxr-x--x. 1 wanghuimin ssh.randd     404 May 31 13:54 run.multSaps3.sh
-rw-r--r--. 1 wanghuimin ssh.randd   17692 May 31 23:18 run.multSaps3.sh.e743526
-rw-r--r--. 1 wanghuimin ssh.randd     544 May 31 23:18 run.multSaps3.sh.o743526
-rw-r-----. 1 wanghuimin ssh.randd      67 Sep  2 18:18 sh.txt
-rw-r-----. 1 wanghuimin ssh.randd    1156 May 31 15:22 tag celltype.tsv
```

ls *.sh* > sh.txt

```
$ ls *.sh* >sh.txt
```

```
$ cat sh.txt
run.multSaps3.sh
run.multSaps3.sh.e743526
run.multSaps3.sh.o743526
```

Pipe and output redirection

- Pipe operator is "|"; it can pass the standard output (stdout) of a previous command to a subsequent command as the standard input (stdin) of a later command.
- Note that it can only pass standard output, not error output (stderr) to subsequent commands.



- command1 | command2
- command1 | command2 [| commandN...]

- Redirection operator ">"; Concatenate a command to a file and use the file to receive the output of the command.

- command > file
- command1 | command1

Linux shell executes multiple commands

1. The commands are separated by ;
   - The execution result of each command does not affect the execution of other commands. That is, all commands are executed, but not all commands are executed successfully.

2. The commands are separated by &&
   - The following command is executed only if the previous command is successfully executed. That is, all commands are executed successfully.

3. The commands are separated by ||
   - || means or, only after a failed to perform the previous command to execute the next command, until the successful execution a command.

Task submission

Task submission - Avoid spending too much time on the command-line interface

1.Direct Task Submission ： nohup [Tasks to submit] &

```
Windrunner 21:42:38 ~
$ conda activate celescope2
(celescope2) Windrunner 22:01:48 ~
$ cd /mnt/sdd/singleron_training_class/resources/celescope_test/rna/shell
(celescope2) Windrunner 22:05:22 /mnt/sdd/singleron_training_class/resources/celescope_test/rna/shell
$ ll
total 8
drwxr-x--- 9 resources resources 4096 Apr  2 15:01 rna_test
-rwxr-x--- 1 resources resources 1914 Apr  2 14:59 rna_test.sh
(celescope2) Windrunner 22:05:28 /mnt/sdd/singleron_training_class/resources/celescope_test/rna/shell
$ nohup sh rna_test.sh &
```

2. Submitted by SGE： qsub [options] [file.sh]

-V : push the current environment variable
-cwd : indicates that the working directory is the current path
-l : Restricts resources. vf=Memory size, in the unit of K/M/G. p=number of cpus. hostnam= Name of the running node.
-q : specifies a queue
-N : specifies the task name

```
qsub -cwd -V -l vf=10g,p=10 test.sh
```

Singleron

04 Conda introduction and Miniconda installation



What is conda?

- Conda is an open-source package management and environment management system for installing multiple versions of software packages and their dependencies and easily switching between them.
- Conda was created for Python programs for Linux, OS X, and Windows, as well as packaging and distribution of other software.

CONDA

python

Package, dependency and environment management for any Language-Python，R，Ruby, Lua，Scala，Java, Javascript，C/C++，FORTRAN and more.

ANACONDA®

website：https://www.anaconda.com/

Why use conda?

In the early days of bioinformatics, software installation was a hassle, with many dependencies to resolve.

- Trying to sudo apt (ubuntu) or sudo yum (centOS): Not an administrator, no permissions!
- Trying to download a binary file: The software does not offer a binary version!
- Trying to recompile: You can't understand the code, you can't find the relevant information, you have installation problems, and you end up with an installation failure!

The appearance of conda finally solved the big problem of software installation!!!

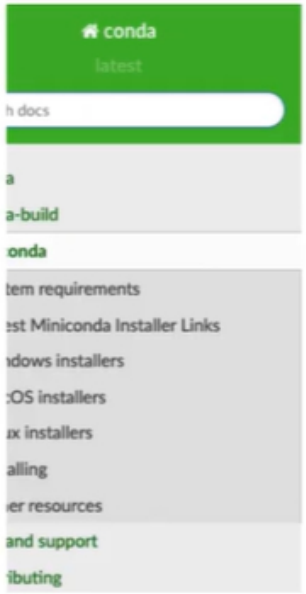


Single cell multi-omics for precision medicine

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Download miniconda and install (If you need it on your own)

MINICONDA



Latest Miniconda Installer Links

Latest - Conda 22.11.1 Python 3.10.8 released December 2

Platform	Name	SHA256 hash
Windows	Miniconda3 Windows 64-bit	2e3086630fa3fae7636432a954be530c88d0705fce497120d56e0f5d865b0d51
	Miniconda3 Windows 32-bit	4fb64e6c9c28b88beab16994bfba4829110ea3145baa60bda5344174ab65d462
macOS	Miniconda3 macOS Intel x86 64-bit bash	7406579393427eaf9bc0e094dcd3c66d1e1b93ee9db4e7686d0a72ea5d7c0ce5
	Miniconda3 macOS Intel x86 64-bit pkg	9195ffba1a6984c81c69649ce976a38455ace5b474c24a4363e5ca65fc72e832
	Miniconda3 macOS Apple M1 64-bit bash	22eec9b7d3add25ac3f9b60621d8f3d8df3e63d4aa0ae5eb846b558d7ba68333
	Miniconda3 macOS Apple M1 64-bit pkg	fbb33c5770b10a0d5a0deef746e7499bfaf8ff840d0d517175036dd8449357f6
Linux	Miniconda3 Linux 64-bit	00938c3534750a0e4069499baf8f4e6dc1c2e471c86a59caa0dd03f4a9269db6
	Miniconda3 Linux-aarch64 64-bit	48a96df9ff56f7421b6dd7f9f71d548023847ba918c3826059918c08326c2017
	Miniconda3 Linux-ppc64le 64-bit	4c86c3383bb27b44f7059336c3a46c34922df42824577b93eadecefbf7423836
	Miniconda3 Linux-s390x 64-bit	a150511e7fd19d07b770f278fb5dd2df4bc24a8f55f06d6274774f209a36c766

- Follow the instructions in the official website: https://conda.io/projects/conda/en/stable/user-guide/install/index.html

- Not required for our BI Course.

Singleron

05 Basic use of conda



Channel?



Download conda

Install conda

Configure the conda channel

Downloading the Installation Package





Buy a television

Mount the TV Set

Search for channels on your TV

Watch TV



Adding channels

- Official channels:

```
conda config --add channels bioconda
conda config --add channels conda-forge
conda config –set show channel_urls yes
```

tip1: Do not add channels repeatedly, redundant channels can slow down the process of installing software. The order makes sense!!
tip2: Adding channels depends on where the server is located, **not where the user is!!**

- **View the added channels in ~/.condarc**

- View the added channels :
  conda config --get channels
  conda config --show channels

```
[ec2-user@ip-10-0-1-246 TCR]$ conda config --get channels
--add channels 'defaults'   # lowest priority
--add channels 'r'
--add channels 'conda-forge'
--add channels 'bioconda'   # highest priority
[ec2-user@ip-10-0-1-246 TCR]$ conda config --show channels
channels:
  - bioconda
  - conda-forge
  - r
  - defaults
```

Create independent environment

- What is the environment?
- Why create an independent environment?
  - conda changes your environment
  - Environmental dependence problem
    Software A depends on python2.7
    Software B depends on python3.8
- Manage different projects
  DNA-seq、RNA-seq、Chip-seq、scRNA-seq、
  ...

- Advice:

1. It is better not to pack any packages in the base environment
2. Unless, you know what the package will do to your native environment.

Create independent environment

- **conda create -n R4**

  Create a conda small environment named R4

  -n: Specifying an environment Name
- **conda activate R4**

  Start the conda environment R4
- **conda deactivate**

  Exit the conda environment R4
- **conda env list**
- **conda info --env**

  List existing microenvironments

```
(base) [Singleron@ecs-23f0 ~]$ conda info --env
# conda environments:
#
base                  *  /mnt/sdd/Singleron/miniconda3
R4                       /mnt/sdd/Singleron/miniconda3/envs/R4
```

```
(base) [Singleron@ecs-23f0 ~]$ conda create --name R4
Collecting package metadata (current_repodata.json): done
Solving environment: done


==> WARNING: A newer version of conda exists. <==
  current version: 4.11.0
  latest version: 4.12.0

Please update conda by running

    $ conda update -n base -c defaults conda



## Package Plan ##

  environment location: /mnt/sdd/Singleron/miniconda3/envs/R4



Proceed ([y]/n)? y

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate R4
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) [Singleron@ecs-23f0 ~]$ conda activate R4
(R4) [Singleron@ecs-23f0 ~]$ conda deactivate
(base) [Singleron@ecs-23f0 ~]$
```

Delete & rename environment and common Conda commands

- Delete the created environment and installed packages

  conda remove -n R4 --all

- How to rename a small environment?

  conda create -n Python2

  conda create -n py2 --clone Python2

  conda remove -n Python2 --all

```
(base) [Singleron@ecs-23f0 ~]$ which conda
~/miniconda3/bin/conda
(base) [Singleron@ecs-23f0 ~]$ pwd
/mnt/sdd/Singleron
(base) [Singleron@ecs-23f0 ~]$ conda --help
usage: conda [-h] [-V] command ...

conda is a tool for managing and deploying applications, environments and packages.

Options:

positional arguments:
  command
    clean       Remove unused packages and caches.
    compare     Compare packages between conda environments.
    config      Modify configuration values in .condarc. This is modeled after the git config command. Writes to the user
                .condarc file (/mnt/sdd/Singleron/.condarc) by default.
    create      Create a new conda environment from a list of specified packages.
    help        Displays a list of available conda commands and their help strings.
    info        Display information about current conda install.
    init        Initialize conda for shell interaction. [Experimental]
    install     Installs a list of packages into a specified conda environment.
    list        List linked packages in a conda environment.
    package     Low-level conda package utility. (EXPERIMENTAL)
    remove      Remove a list of packages from a specified conda environment.
    uninstall   Alias for conda remove.
    run         Run an executable in a conda environment. [Experimental]
    search      Search for packages and display associated information. The input is a MatchSpec, a query language for
                conda packages. See examples below.
    update      Updates conda packages to the latest compatible version.
    upgrade     Alias for conda update.

optional arguments:
  -h, --help     Show this help message and exit.
  -V, --version  Show the conda version number and exit.

conda commands available from other packages:
  content-trust
  env
```

- conda-Use of help documents

  conda --help

- conda common usage options

  conda config Channel address

  conda search Software name

  conda install Software name

  conda uninstall Software name

  conda remove package name

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Common Installation Software

- QC : fastqc、miltiqc、fastp、trimmomatic、cutadapt、trim_galor...
- Transcriptome alignment & quantification : bwa、hisat2、bowtie、bowtie2、STAR、salmon、subread (featureCount) ....
- Assembly of genome：Trinity
- Single cell transcriptome：CeleScope
- ……

Hands-on exercise

1- Use wget to download linux practice data to your own linux system and decompress it

wget https://singleronbio-opendata.oss-cn-hangzhou.aliyuncs.com/Online_training_Materials/linux/linux_traindata.tar.gz

- Add '--no-check-certificate' option if fails

2- Create a directory (folder) named linux_test in your practice path

3- Copy the catfile.txt and dog.txt files to the linux_test folder with **only one line** of code

4- View the first six lines of the dog.txt file

5- Merge files catfile.txt and dog.txt into one file named newfile.txt

6- List the files containing the string "file" in file name inside linux_test folder and write the output to test.txt

7- Create a folder named test2

8- Copy the gene folder to the linux_test directory, rename it GENE, and compress it into tar.gz

9- Move the compressed GENE file to test2 and extract it

10- Change the folder permission of linux_test to read, write, and execute by all users.